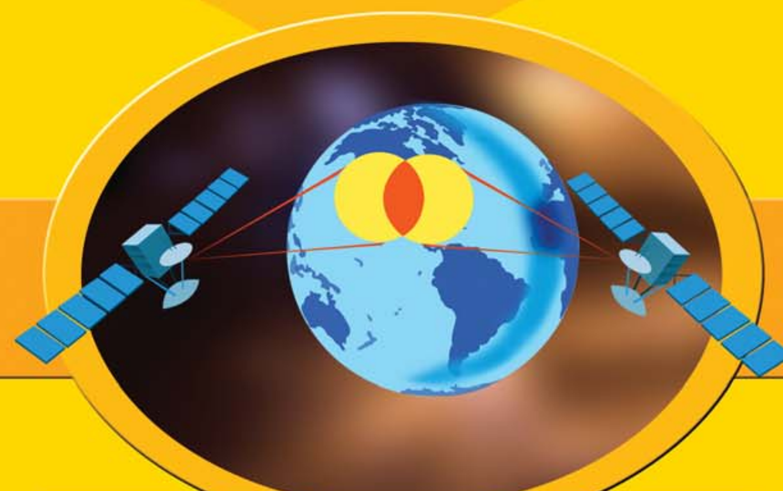
How GPS Works: From The GPS Constellation To Differential GPS

Twenty-seven satellites orbiting the earth are responsible for the location-tracking information any GPS receiver—be it in a car or a ship—receives. Of these, at least 24 are operational at any time, and the remaining ones are used as backups.

These satellites continuously send out strings of bits that synchronise with every active GPS receiver. They also send out information about where they are.

Thus, a GPS receiver always has information about the location of satellites within its range.



I. With Only One Satellite

When your GPS receiver gets a signal from one satellite, it knows where the satellite is, and can estimate the distance between itself and the satellite. The receiver thus knows it is located somewhere in the yellow circle (as shown in the picture above).

II. When Two Satellites Come In

If the receiver can get information from two satellites, it can further refine its location: it now knows that it is located somewhere within the red area—the intersection of the two yellow areas (as shown in illustration above).